

LECTURE 04 · REVIEW NOTES

Asset Allocation and Portfolio Optimization

From two assets to the efficient frontier, the Sharpe ratio, the risk-free asset and the tangency portfolio.

About these notes

These notes review the material of Lecture 04. Each concept is explained in full, and each numerical example is worked out step by step, with every intermediate figure shown. The graded activities completed during class are summarized in general terms and marked as in-class activities.

1. The two-asset frontier and the role of correlation

The construction of an optimal portfolio starts with the simplest case: two risky assets. Throughout this section, asset A is a stock fund with $E[r] = 12\%$ and $\sigma = 20\%$, and asset B is a bond fund with $E[r] = 4\%$ and $\sigma = 10\%$. For any weights w_A and $w_B = 1 - w_A$:

$$E[r_p] = w_A E[r_A] + w_B E[r_B]$$

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho$$

The expected return of any mix is a weighted average and does not depend on the correlation ρ . The variance does: the last term, which contains ρ , determines how far the portfolio's risk falls below the weighted average of the two risks.

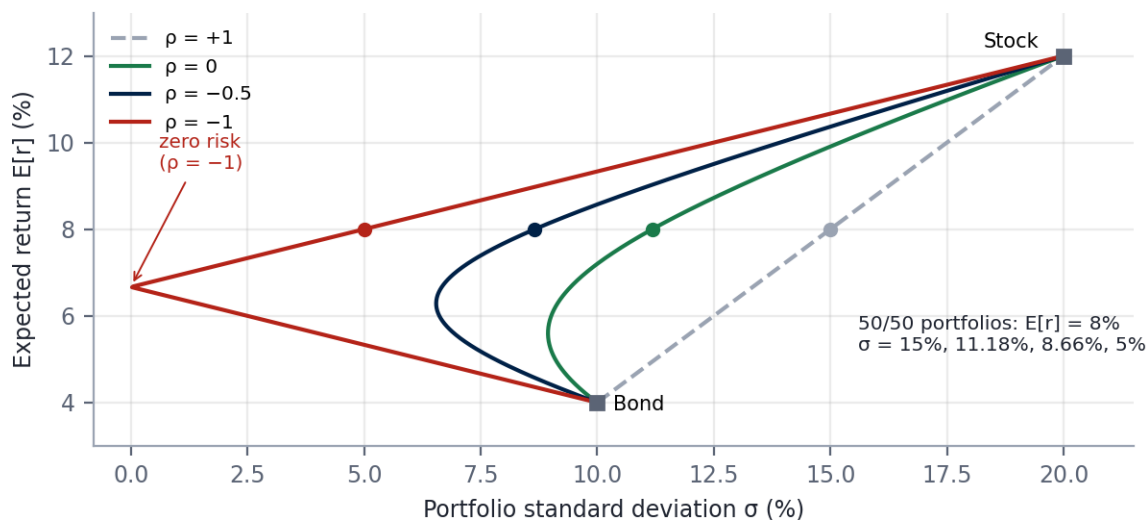
WORKED EXAMPLE 1

A 50/50 portfolio under four correlations

With $w_A = w_B = 0.50$, the expected return is $0.50 \times 12\% + 0.50 \times 4\% = 8\%$ in every case. The two own-variance terms are also fixed: $(0.50)^2(0.20)^2 = 0.0100$ and $(0.50)^2(0.10)^2 = 0.0025$, which sum to 0.0125. The cross-term is $2(0.50)(0.50)(0.20)(0.10) \times \rho = 0.0100 \times \rho$.

Correlation ρ	Cross-term $0.0100 \times \rho$	Variance σ_p^2	σ_p	$E[r_p]$
+1	+0.0100	$0.0125 + 0.0100 = 0.0225$	$\sqrt{0.0225} = 15.00\%$	8%
0	0	$0.0125 + 0 = 0.0125$	$\sqrt{0.0125} = 11.18\%$	8%
-0.5	-0.0050	$0.0125 - 0.0050 = 0.0075$	$\sqrt{0.0075} = 8.66\%$	8%
-1	-0.0100	$0.0125 - 0.0100 = 0.0025$	$\sqrt{0.0025} = 5.00\%$	8%

With $\rho = +1$ the portfolio's risk is exactly the weighted average, $0.5 \times 20\% + 0.5 \times 10\% = 15\%$, and the possible portfolios lie on a straight line between the two assets: there is no diversification benefit. As ρ falls, the curve bends to the left. At $\rho = 0$ the same 50/50 mix has 11.18% risk, at $\rho = -0.5$ it has 8.66%, and at $\rho = -1$ only 5%. The expected return never changes.



All combinations of the stock and bond funds for four correlations. The dots mark the 50/50 portfolio in each case, which always has an expected return of 8%.

WORKED EXAMPLE 2**The minimum-variance portfolio**

As the curve bends, one portfolio has the lowest possible risk: the **minimum-variance portfolio** (MVP). For two assets its weight on A is:

$$w_A^* = (\sigma_B^2 - \sigma_A \sigma_B \rho) \div (\sigma_A^2 + \sigma_B^2 - 2\sigma_A \sigma_B \rho)$$

Step	Calculation	Result
1	$\rho = -0.5$: covariance $\sigma_A \sigma_B \rho = 0.20 \times 0.10 \times (-0.5)$	-0.0100
2	Numerator = $0.0100 - (-0.0100)$	0.0200
3	Denominator = $0.0400 + 0.0100 - 2(-0.0100)$	0.0700
4	$w_A^* = 0.0200 \div 0.0700$	28.6% stock, 71.4% bond
5	$E[r] = 0.286 \times 12\% + 0.714 \times 4\%$	6.29%
6	$\sigma^2 = 0.286^2(0.04) + 0.714^2(0.01) + 2(0.286)(0.714)(-0.01) = 0.003265 + 0.005102 - 0.004082$; $\sigma = \sqrt{0.004285}$	$\sigma = 6.55\%$
7	$\rho = -1$: $w_A = \sigma_B \div (\sigma_A + \sigma_B) = 10 \div 30$; $E[r] = \frac{1}{3}(12\%) + \frac{2}{3}(4\%)$	33.3% stock, $E[r] = 6.67\%$, $\sigma = 0$

At $\rho = -1$ the two assets hedge each other perfectly, and a portfolio with one third in stocks has no risk at all. Only the part of each curve above the MVP is efficient: every portfolio below it has a twin above it with the same risk and a higher return.

IN-CLASS ACTIVITY · How correlation shapes the frontier

An interactive tool in class let the correlation and the weight on the stock fund be varied continuously, showing the live variance calculation and how the curve bends further left as ρ falls.

2. From two assets to N assets

Real allocation problems involve many assets. The lecture used the eleven S&P 500 sector ETFs, with annualized statistics estimated from monthly returns between January 2020 and August 2026:

ETF	Sector	E[r]	σ	ETF	Sector	E[r]	σ
XLK	Technology	15.0%	28.5%	XLY	Consumer discretionary	9.0%	28.1%
XLF	Financials	9.0%	26.3%	XLP	Consumer staples	5.5%	11.4%
XLE	Energy	8.0%	39.9%	XLB	Materials	9.0%	24.4%
XLV	Health care	8.0%	15.7%	XLR E	Real estate	1.5%	23.4%
XLI	Industrials	9.0%	23.3%	XLU	Utilities	0.5%	18.0%
XLC	Communication	10.0%	24.8%				

With two assets every portfolio lies on one curve. With eleven, the possible portfolios fill a whole region of the $(\sigma, E[r])$ plane, and the individual ETFs sit inside or to the right of its boundary. The number of inputs also grows quickly: eleven expected returns, eleven variances and $11 \times 10 \div 2 = 55$ distinct covariances, for 66 distinct entries in the covariance matrix. For N assets the covariance matrix has $N(N + 1) \div 2$ distinct entries, which is why estimation error becomes a central practical problem in large portfolios.

IN-CLASS ACTIVITY · Part I: compute the return and risk of an assigned portfolio

Each student received a set of weights across the eleven ETFs and a spreadsheet of monthly returns, and used it to compute the portfolio's expected return and standard deviation. The submitted portfolios were plotted together, showing how widely risk and return vary across allocations of the same eleven assets.

3. Finding the efficient frontier: an optimization problem

The **efficient frontier** is the set of portfolios with the highest expected return for each level of risk. Finding it is an optimization problem that can be written in two equivalent ways. Using w for the vector of weights, μ for the vector of expected returns and Σ for the covariance matrix:

	① Maximize return for a given risk	② Minimize risk for a given return
Objective	$\max w' \mu$	$\min w' \Sigma w$
Risk or return constraint	$w' \Sigma w \leq \bar{\sigma}^2$ (risk budget)	$w' \mu \geq \bar{r}$ (return target)
Fully invested	$\Sigma w_i = 1$	$\Sigma w_i = 1$
Long-only	$w_i \geq 0$	$w_i \geq 0$
Tracing the frontier	Repeat for $\bar{\sigma}$ from σ_{MVP} upward	Repeat for $\bar{r}$ from the minimum to the maximum return

Both formulations trace exactly the same curve. Because the portfolio variance $w'\Sigma w$ is a quadratic function of the weights, the problem is a **quadratic program** (QP), which standard tools solve reliably: Excel Solver, Python's `scipy` or `CVXPY`, or any commercial optimizer. The long-only constraint $w_i \geq 0$ rules out short positions; Lecture 05 relaxes it.

IN-CLASS ACTIVITY · Part II: find the optimal portfolio for a target risk

Each student received a target standard deviation and used Excel Solver to find the allocation across the eleven ETFs with the highest expected return at that risk, then compared the answer with the efficient frontier computed for the whole class.

4. The Sharpe ratio

Comparing portfolios with different levels of risk requires a measure of return per unit of risk. The **Sharpe ratio** is the portfolio's expected excess return over the risk-free rate, divided by its standard deviation:

$$SR = (E[r_p] - r_f) \div \sigma_p$$

A higher Sharpe ratio means the investor is better compensated for each unit of risk taken. As a rough guide, a ratio below zero means the portfolio earns less than the risk-free rate, 0 to 0.25 is poor compensation, 0.25 to 0.50 is decent, and above 0.50 is excellent and rare in practice.

WORKED EXAMPLE 3

Sharpe ratios with $r_f = 4.3\%$

Portfolio	Calculation	Sharpe ratio
XLK	Excess return = $15.0\% - 4.3\% = 10.7\%$; $SR = 10.7 \div 28.5$	0.375
XLV	Excess return = $8.0\% - 4.3\% = 3.7\%$; $SR = 3.7 \div 15.7$	0.236
MVP	Minimum-variance portfolio of the 11 ETFs ($E[r] = 6.0\%$, $\sigma = 10.7\%$): $SR = 1.7 \div 10.7$	0.159

XLK has more than twice the risk of XLV, yet it delivers more excess return per unit of risk. The minimum-variance portfolio has the least risk of all, but also the lowest Sharpe ratio of the three: the safest portfolio is not the most efficient use of risk.

5. The risk-free asset and the Capital Allocation Line

The **risk-free rate** is the return on an asset with no default risk and no volatility ($\sigma = 0$). Common proxies are the 3-month U.S. Treasury bill, the overnight SOFR rate and the 1-year Treasury note; the rate used in the lecture was $r_f = 4.3\%$ (September 2026).

Adding a risk-free asset changes the problem in an important way. An investor who puts a fraction y in a risky portfolio P and $1 - y$ in the risk-free asset obtains:

$$E[r_C] = r_f + y \times (E[r_P] - r_f) \text{ and } \sigma_C = y \times \sigma_P$$

Because the risk-free asset has no volatility and no correlation with anything, both return and risk are linear in y . All such combinations lie on a straight line from r_f through P, the **Capital Allocation Line** (CAL), and the slope of that line is exactly the Sharpe ratio of P. The investor therefore no longer picks a point on the curved frontier: the investor picks a risky portfolio and then slides along its CAL.

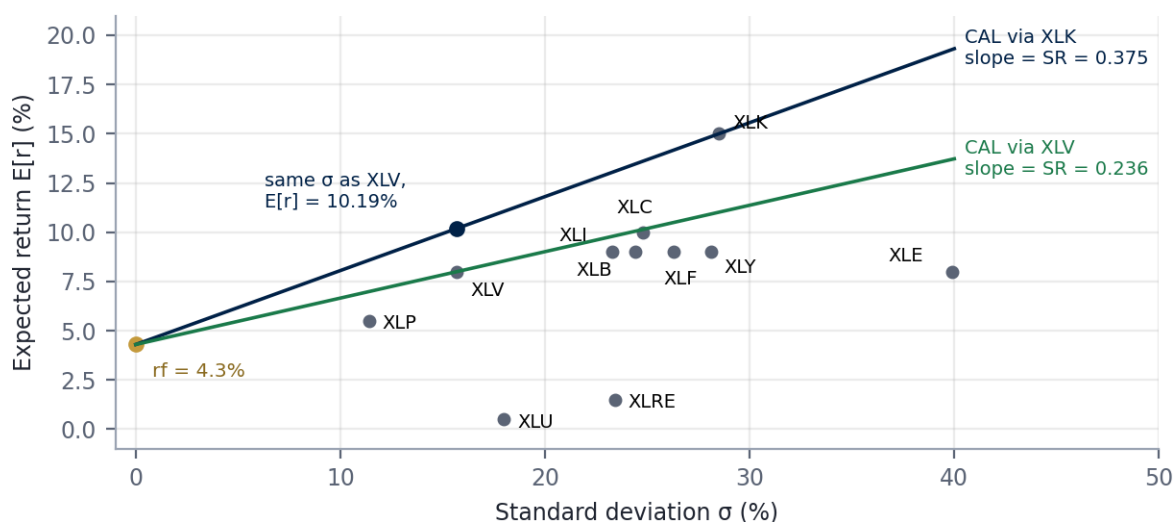
WORKED EXAMPLE 4

Why a higher Sharpe ratio dominates

Suppose an investor wants the same risk as XLV, $\sigma = 15.7\%$, but builds it by combining XLK with the risk-free asset instead of holding XLV.

Step	Calculation	Result
1	Fraction in XLK needed for $\sigma = 15.7\%$: $y = 15.7 \div 28.5$	y = 55.1%
2	Fraction in the risk-free asset = $1 - 0.551$	44.9%
3	$E[r] = 4.3\% + 0.551 \times (15.0\% - 4.3\%) = 4.3\% + 0.551 \times 10.7\%$	10.19%
4	Compare with holding XLV directly at the same σ	E[r] = 8.0%

At identical risk, the XLK-plus-T-bills combination earns 2.19 percentage points more than XLV. This is the Sharpe ratio at work: a steeper CAL offers more return at *every* level of risk, so the investor should always prefer the risky portfolio whose CAL is steepest.



The eleven sector ETFs and two capital allocation lines from $r_f = 4.3\%$. Every point on the steeper line through XLK beats the corresponding point on the line through XLV.

6. The tangency portfolio

Among all portfolios on the efficient frontier, one produces the steepest possible CAL: the line from r_f that just touches the frontier. That point is the **tangency portfolio**, and it is the portfolio with the maximum Sharpe ratio available from the assets under consideration. Lines through portfolios lower on the frontier are flatter; lines through points above the frontier are steeper, but those portfolios cannot be built from the available assets.

The result splits the allocation decision in two. First, every investor who agrees on the inputs should hold the **same risky portfolio**, the tangency portfolio, because it offers the best trade-off. Second, each investor chooses **how much** to put in it, according to personal risk tolerance: a cautious investor holds part tangency and part risk-free asset, while a more aggressive one holds more of the tangency portfolio. Most investors hold something between the risk-free asset and the tangency portfolio. Real portfolios often have lower Sharpe ratios than the tangency portfolio because of equal weighting, constraints or other preferences.

IN-CLASS ACTIVITY · Part III: compare Sharpe ratios

Each student computed the Sharpe ratio of the portfolio from Part I, using $r_f = 4.3\%$, and compared it with the maximum Sharpe ratio attainable with the eleven ETFs.

7. Key takeaways

Diversification helps whenever correlation is below +1. The expected return of a portfolio is a weighted average, but its risk falls below the weighted average as correlation falls, and a perfectly negatively correlated pair can remove risk entirely.

The efficient frontier is the result of an optimization. Maximizing return for a given risk and minimizing risk for a given return trace the same curve, and both are quadratic programs solved with tools such as Excel Solver.

The Sharpe ratio measures return per unit of risk. It is the slope of the capital allocation line, and a steeper line dominates at every level of risk.

The risk-free asset turns the frontier into a line. Combining a risky portfolio with T-bills makes both return and risk linear in the fraction invested.

The tangency portfolio maximizes the Sharpe ratio. Investors should hold that risky portfolio and adjust their overall risk by mixing it with the risk-free asset.

ETF statistics and the risk-free rate are those presented in Lecture 04. The minimum-variance calculations in Example 2 and the comparison in Example 4 apply the lecture's figures.